



Statistics Department
Arabic Section
Fourth Level



Categorical project

“Determinants of Happiness in the United States, 2017”

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Introduction:

In the first part of this project, we have provided a comprehensive descriptive and bivariate examination of happiness in the United States using data from the World Values Survey (Wave 7). Through frequency distributions, contingency tables, and chi-square-based association tests, the analysis offered an initial understanding of how individual attitudes and socio-demographic characteristics relate to self-reported happiness. While these methods are essential for identifying patterns and preliminary relationships, they remain limited in their ability to account for the simultaneous influence of multiple factors. The descriptive analysis revealed notable variation in happiness levels across key variables. In particular, individuals reporting higher levels of economic satisfaction were consistently more likely to classify themselves as happy. Similarly, respondents who attached greater importance to religion and those who expressed stronger respect for authority tended to report higher happiness levels. These patterns were observed repeatedly across the contingency tables and supported by statistically significant chi-square tests. The analysis now shifts from descriptives to modeling. Given the nature of the dependent variable, binary logistic regression is the most appropriate modeling strategy.

Model Specification:

The dependent variable in the logistic regression model is Happiness, operationalized as a binary outcome. Following the descriptive and bivariate analysis conducted in Part 1, the original categorical happiness variable was recoded into a dichotomous indicator suitable for binary logistic regression.

The model includes a combination of attitudinal and socio-demographic variables that were motivated by both theoretical considerations and empirical findings from the bivariate analysis. The full model specification incorporates the following explanatory variables:

- Importance of religion, capturing the subjective role of religion in the respondent's life
- Tolerance, reflecting attitudes toward social diversity
- Respect for authority, representing attitudes toward hierarchical and institutional norms

- Economic satisfaction, measuring perceived satisfaction with personal economic conditions
- Gender, included as a basic demographic control
- Age group, accounting for life-cycle effects
- Region, distinguishing between different areas of residence
- Education, included in recoded form to ensure an appropriate functional relationship

All categorical predictors were entered into the model using dummy-variable coding, with one category designated as the reference group. Reference categories correspond to the first level of each factor. The reference categories used in the model are as follows:

- Religion importance: Important (reference)
- Tolerance: Yes (reference)
- Respect for authority: Not interested (reference)
- Economic satisfaction: Dissatisfied (reference)
- Gender: Male (reference)
- Age group: 16–29 years (reference)
- Region: Urban (reference)
- Education: Lower/Middle education (reference)

Analysis:

Prior to model estimation, the dataset was prepared for binary logistic regression by recoding the original happiness variable into a dichotomous outcome. Respondents who reported being happy were coded as 1, while those reporting otherwise were coded as 0. This transformation ensures compatibility with the binary logit.

As we already know the binary happiness variable is highly imbalanced. About 87.5% of the respondents are classified as happy(1), while only 12.5% are

```
> table(df_model$happiness_binary)
 0    1
315 2201
> prop.table(table(df_model$happiness_binary))
      0      1
0.1251987 0.8748013
```

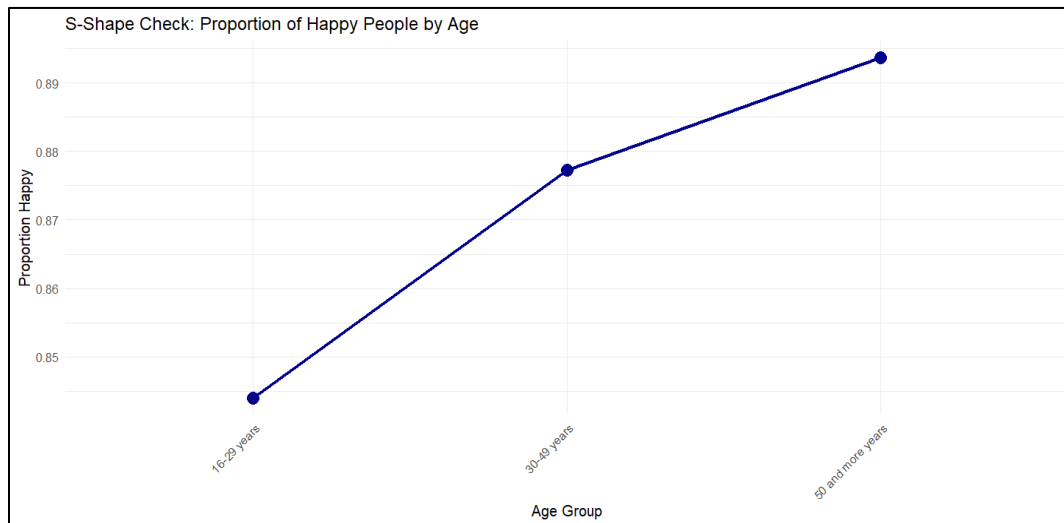
classified as not happy(0).which is important to consider in the modeling stage.

Checking Assumptions of Logistic Regression:

In this section, we will check the key assumptions of logistic regression. In particular, we focus on the assumption of a logistic (S-shaped) relationship between the predictors and the log-odds of the outcome. For variables with more than two categories, we examine whether the proportion of “happy” respondents increases or decreases in an S-shaped pattern across the levels of the predictor. This helps us assess whether the logistic form is appropriate.

For binary predictors, the S-shape assumption cannot be visually evaluated, because binary variables only have two possible values, which produce only two points rather than a curve. Therefore, for binary variables we simply compare the proportion of happy versus not happy respondents across the two categories instead of checking for an S-shaped pattern.

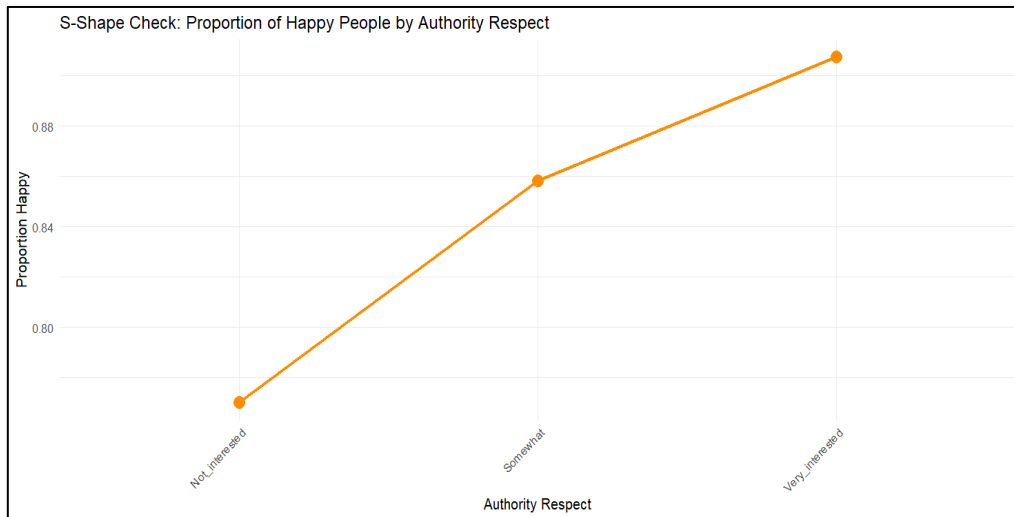
I. Age:



To examine whether the logistic functional form is appropriate for age, we plotted the proportion of respondents who report being happy across the three age groups. The graph shows a clear monotonic increase in the proportion of happy individuals as age increases. Happiness is lowest among respondents aged 16–29, increases among those aged 30–49, and reaches its highest level among respondents aged 50 and above. Although the pattern

is not perfectly S-shaped, it is consistent with a smooth, non-linear relationship between age and the log-odds of happiness, which supports the use of a logistic specification. Importantly, there is no evidence of an erratic or U-shaped pattern that would contradict the logistic regression assumption.

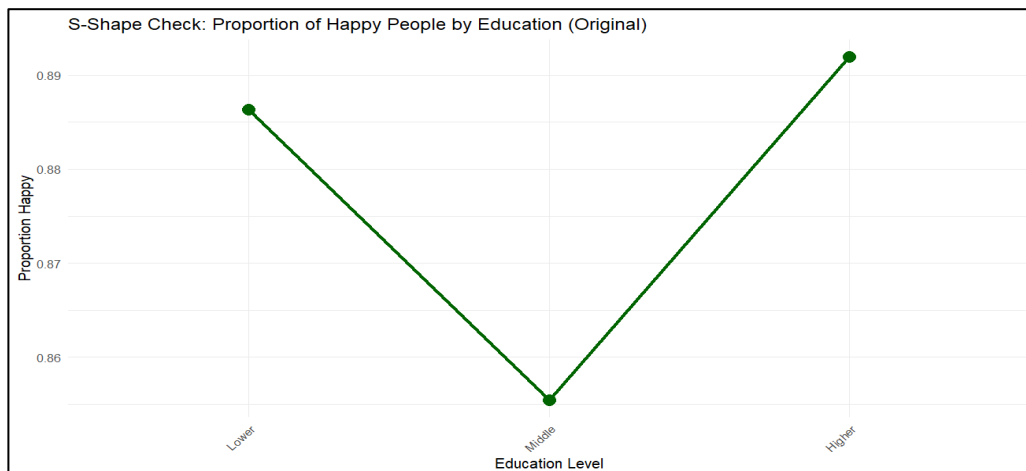
II. Authority Respect:



diagnostic was conducted for authority respect. The proportion of individuals reporting happiness rises steadily as respect for authority increases. Respondents who are not interested in respecting authority exhibit the lowest levels of happiness, while those who express strong respect show the highest proportions of happiness.

This ordered and monotonic pattern is consistent with the expected logistic relationship and provides no indication of a violation of the S-shape assumption. Consequently, respect for authority can be appropriately modeled using dummy variables without concerns about functional form misspecification.

III. Education:



In contrast to the previous variables, the relationship between education and happiness does not follow a monotonic pattern. As shown in the plot, the proportion of happy respondents is relatively high among individuals with lower education, declines for those with middle education, and then increases again for respondents with higher education. This U-shaped pattern violates the S-shape assumption underlying logistic regression, indicating that the original coding of education does not provide an appropriate functional form.

In response to this pattern, education was recoded by combining the lower and middle education categories into a single group and contrasting them with respondents who attained higher education. This recoding transform education into a binary predictor. As a result, an S-shape assessment is no longer applicable, since such a functional form is not meaningful for binary variables. Consequently, the relationship between the recoded education variable and happiness is examined using proportion tables rather than graphical S-shape diagnostics.

	education_binary	n_total	n_happy	prop_happy
	<fct>	<int>	<int>	<dbl>
1	Lower/Middle	605	516	0.853
2	Higher	1911	1685	0.882

Checking Proportions for Binary Variables:

After examining the S-shape patterns for ordinal variables, we now assess the proportions of happiness across the binary predictors. For each binary variable, we calculate the proportion of happy individuals in each category to identify potential associations with happiness before including them in the logistic regression model.

1. Religion Importance:

The proportion of happy individuals is notably higher among those who consider religion important (90.8%) compared to those who do not (83.1%), showing a difference of approximately 7.7 percentage points. This suggests that religion importance may have a meaningful association with happiness and could be a relevant predictor in our logistic regression model.

```
> print(prop_religion)
# A tibble: 2 x 4
```

	religion	n_total	n_happy	prop_happy
	<fct>	<int>	<int>	<dbl>
1	Important	1432	1300	0.908
2	Not Important	1084	901	0.831

2. Gender:

The proportions of happiness are very similar between males (87.1%) and females (87.9%), with only a 0.8 percentage point difference. This minimal difference suggests that gender may not be a strong predictor of happiness in our sample, though we will retain it in the model to control for potential confounding effects.

```
> print(prop_gender)
# A tibble: 2 x 4
```

	gender	n_total	n_happy	prop_happy
	<fct>	<int>	<int>	<dbl>
1	M	1356	1181	0.871
2	F	1160	1020	0.879

3.Region:

Urban and rural residents show nearly identical happiness proportions (87.5% and 87.6%, respectively), with virtually no difference between the two groups. This indicates that region of residence may not significantly influence happiness levels in our dataset, though it will be included in the full model for completeness.

```
> print(prop_region)
# A tibble: 2 x 4
  region n_total n_happy prop_happy
<fct>   <int>   <int>   <dbl>
1 Urban    2234    1954    0.875
2 Rural     282     247    0.876
```

4.Tolerance:

Individuals who answered "Yes" to tolerance have a slightly higher proportion of happiness (88.2%) compared to those who answered "No" (85.8%), with a difference of 2.4 percentage points. While this difference is modest, it suggests that tolerance may have some association with happiness.

```
> print(prop_tolerance)
# A tibble: 2 x 4
  tolerance n_total n_happy prop_happy
<fct>      <int>   <int>   <dbl>
1 Yes       1760    1552    0.882
2 No         756     649    0.858
```

5.Economic Satisfaction:

Economic satisfaction shows the strongest association with happiness among all binary variables. Individuals who are satisfied economically have a substantially higher proportion of happiness (94.1%) compared to those who are dissatisfied (76.9%), representing a difference of 17.2 percentage points. This large difference indicates that economic satisfaction is likely to be a highly significant predictor of happiness in our logistic regression model.

```
> print(prop_economic)
# A tibble: 2 x 4
  economical_satisfaction n_total n_happy prop_happy
<fct>                   <int>   <int>   <dbl>
1 Dissatisfied           965     742    0.769
2 Satisfied             1551    1459    0.941
```

Binary Logistic Regression Model:

Model Specification

To investigate the determinants of individual happiness, a binary logistic regression model was estimated. The dependent variable is binary, taking the value 1 if the individual is classified as happy and 0 otherwise. Given the binary nature of the outcome variable, the logit specification was employed to model the probability of being happy as a function of social, economic, and demographic characteristics.

The model is specified as follows:

$$\log\left(\frac{\pi(\text{Happiness}=1)}{1-\pi(\text{Happiness}=1)}\right) = \beta_0 + \beta_1 \text{Religion}_i + \beta_2 \text{Tolerance}_i + \beta_3 \text{AuthorityRespect}_i^{(L)} + \beta_4 \text{AuthorityRespect}_i^{(Q)} + \beta_5 \text{EconomicSatisfaction}_i + \beta_6 \text{Gender}_i + \beta_7 \text{Age}_i^{(L)} + \beta_8 \text{Age}_i^{(Q)} + \beta_9 \text{Region}_i + \beta_{10} \text{Education}_i$$

where $\pi(\text{Happiness} = 1)$ denotes the probability that individual i is happy. The coefficients represent the marginal effects of each explanatory variable on the log-odds of happiness, holding all other variables constant. Ordinal variables (authority respect and age) were included using polynomial contrasts to capture potential non-linear relationships. The estimation results reveal several statistically significant predictors of happiness.

Religion Importance:

Religion importance is a strong and statistically significant determinant of happiness. Individuals who report that religion is *not important* have significantly lower odds of being happy compared to those who consider religion important ($\beta = -0.612$, $p < 0.001$). This negative association persists even after controlling for economic satisfaction and demographic factors.

Tolerance:

Tolerance exhibits a statistically significant relationship with happiness. Respondents who reported being non-tolerant show lower odds of happiness relative to tolerant individuals ($\beta = -0.368$, $p = 0.008$), indicating that tolerance is independently associated with higher happiness levels.

Authority Respect:

Authority respect was modeled as an ordinal variable using polynomial contrasts. The linear component is positive and highly significant ($\beta = 0.645$, $p < 0.001$), suggesting that higher respect for authority is associated with increased odds of happiness. The quadratic component is statistically insignificant ($p = 0.49$), indicating no evidence of a non-linear effect.

Economic Satisfaction:

Economic satisfaction emerges as the most influential predictor in the

```
> summary(logit_model_full)

Call:
glm(formula = happiness_binary ~ religion + tolerance + authority_respect +
     economical_satisfaction + gender + age + region + education_binary,
     family = binomial(link = "logit"), data = df_model)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)      1.131117   0.182233   6.207 5.40e-10 ***
religionNot Important -0.611759   0.136665  -4.476 7.59e-06 ***
toleranceNo       -0.368067   0.138017  -2.667  0.00766 **
authority_respect.L  0.645348   0.128413   5.026 5.02e-07 ***
authority_respect.Q -0.077190   0.111880  -0.690  0.49024
economical_satisfactionSatisfied 1.588029   0.135429  11.726 < 2e-16 ***
genderF           0.150363   0.134683   1.116  0.26424
age.L             0.119193   0.123522   0.965  0.33457
age.Q            -0.142456   0.105693  -1.348  0.17771
regionRural       0.005941   0.203273   0.029  0.97668
education_binaryHigher 0.321582   0.145095   2.216  0.02667 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 1897.9  on 2515  degrees of freedom
Residual deviance: 1667.9  on 2505  degrees of freedom
AIC: 1689.9

Number of Fisher Scoring iterations: 5
```

model. Economically satisfied individuals have substantially higher odds of being happy compared to dissatisfied individuals ($\beta = 1.588$, $p < 0.001$). This result highlights the dominant role of perceived economic well-being in explaining happiness.

Gender:

The effect of gender is not statistically significant ($\beta = 0.150$, $p = 0.264$), indicating that there is no meaningful difference in happiness between males and females once other factors are controlled for.

Age:

Age was entered using polynomial contrasts. Neither the linear nor the quadratic component is statistically significant ($p = 0.335$ and $p = 0.178$, respectively), suggesting that age does not exert an independent effect on happiness in this specification.

Region:

Region of residence does not significantly influence happiness. The coefficient for rural residence is close to zero and statistically insignificant ($\beta = 0.006$, $p = 0.977$), implying no observable difference between urban and rural respondents.

Education:

Higher education is positively and significantly associated with happiness ($\beta = 0.322$, $p = 0.027$). Individuals with higher educational attainment have greater odds of being happy compared to those with lower education levels.

Model Fit and Overall Assessment:

The fitted model represents a substantial improvement over the null model, as indicated by the reduction in deviance from 1897.9 to 1667.9. The Akaike Information Criterion (AIC) value of 1689.9 suggests a satisfactory balance between model fit and complexity. Overall, the results indicate that economic satisfaction, religion importance, authority respect, tolerance, and education are key determinants of happiness, while gender, age, and region do not have statistically significant independent effects.

Model Diagnostics and Goodness of Fit:

After fitting the binary logistic regression model, it is essential to evaluate how well the model explains the observed data and whether it provides an adequate representation of the relationship between happiness and the set of explanatory variables.

i. Using pseudo R-squared measures:

```
> print(pseudo_r2)
```

11h	11hNull	G2	McFadden	r2ML	r2CU
-833.93811699	-948.92622677	229.97621956	0.12117708	0.08735244	0.16491968

Log-likelihood of the fitted model (llh): -833.94

This value reflects how well the estimated logistic regression model fits the observed data. A **higher (less negative)** log-likelihood indicates a better fit. Compared with the null model, this value shows a **substantial improvement in model fit**, meaning the included predictors meaningfully explain happiness.

Log-likelihood of the null model (llhNull): -948.93

Represents the fit of a model containing **only the intercept**, with no explanatory variables. The large difference between llhNull and llh indicates that **adding predictors significantly improves explanatory power**. This confirms that happiness is **not random**, but systematically related to the included covariates.

Likelihood ratio chi-square (G2): 229.98

Measures the improvement of the full model relative to the null model. A large G^2 value indicates a strong improvement in fit. With $p < 0.001$, the full model is statistically superior to the null model. This provides formal evidence that the predictors jointly contribute to explaining happiness.

McFadden's pseudo R^2 : 0.121

Compares the log-likelihood of the full model to that of the null model. A value of **0.121 indicates a meaningful model fit**. This suggests that the model captures a **non-trivial proportion of happiness variation**, despite the complexity of human well-being.

- The pseudo R-squared measures indicate that the logistic regression model provides a statistically significant and substantively meaningful explanation of happiness, with explanatory power that is typical and acceptable for social science survey data.

ii. Using Hosmer-Lemeshow Goodness of Fit Test:

The Hosmer–Lemeshow (H–L) goodness-of-fit test is used to assess how well a logistic regression model fits the observed data. It compares the **observed outcomes** with the **predicted probabilities** from the model by grouping observations into deciles of predicted risk. A good-fitting model shows no significant difference between observed and expected frequencies.

- **Null hypothesis (H_0):**

The logistic regression model fits the data well; there is no significant difference between observed and predicted values.

- **Alternative hypothesis (H_1):**

The logistic regression model does not fit the data well; there is a significant difference between observed and predicted values.

```
Hosmer and Lemeshow goodness of fit (GOF) test  
data: df_model$happiness_binary, fitted(logit_model_full)  
X-squared = 6.0324, df = 8, p-value = 0.6436
```

Since **Chi-square (χ^2)**: $6.03 < \chi^2 (0.5, df = 8)$, $p\text{-value} = 0.6439 > 0.05$, rejecting the null hypothesis at 5% significance level, indicating that there is **no evidence of poor fit**. The observed proportions of happiness are **consistent with the model's predicted probabilities** across risk groups.

- The Hosmer–Lemeshow goodness-of-fit test indicates that the logistic regression model fits the data adequately ($\chi^2 = 6.03$, $df = 8$, $p = 0.64$). The high p-value suggests no significant discrepancy between observed and predicted outcomes, confirming that the model provides a satisfactory fit to the data

iii. Using Likelihood Ratio Test

The Likelihood Ratio Test (LRT) is used to assess whether the full logistic regression model provides a significantly better fit to the data than the null model (intercept-only). It evaluates whether the set of explanatory variables, taken together, significantly improves the model's ability to explain the outcome variable.

Null hypothesis (H_0):

All regression coefficients (except the intercept) are equal to zero; the full model does not improve fit compared to the null model.

Alternative hypothesis (H_1):

At least one regression coefficient is different from zero; the full model provides a better fit than the null model.

```
Deviance difference: 229.9762  
> cat("DF difference:", df_diff, "\n")  
DF difference: 10  
> cat("P-value:", p_value_lrt, "\n")  
P-value: 0
```

The Likelihood Ratio Test strongly rejects the null hypothesis ($\chi^2 = 229.98$, $df = 10$, $p < 0.05$), indicating that the full logistic regression model fits the data significantly better than

the intercept-only model. This result confirms that the explanatory variables jointly contribute to explaining variation in happiness.

iv. Analysis of Deviance

The Analysis of Deviance table for a logistic regression model provides sequential likelihood ratio tests. Each predictor is added to the model one at a time, and the test evaluates whether the inclusion of that variable significantly improves the model fit by reducing the residual deviance.

- **Null hypothesis (H_0):**
Adding the predictor does not improve model fit; its coefficient(s) are equal to zero.
- **Alternative hypothesis (H_1):**
Adding the predictor significantly improves model fit; at least one coefficient differs from zero.

Null Model:

Residual deviance = 1897.8 (df = 2515). Represents the baseline model with no predictors.

Analysis of Deviance Table						
Model: binomial, link: logit						
Response: happiness_binary						
Terms added sequentially (first to last)						
		Df	Deviance	Resid. Df	Resid. Dev	Pr(>Chi)
NULL				2515	1897.8	
religion	1	32.738		2514	1865.1	1.055e-08 ***
tolerance	1	5.564		2513	1859.5	0.01833 *
authority_respect	2	27.318		2511	1832.2	1.169e-06 ***
economical_satisfaction	1	156.524		2510	1675.7	< 2.2e-16 ***
gender	1	0.649		2509	1675.1	0.42047
age	2	2.380		2507	1672.7	0.30424
region	1	0.011		2506	1672.7	0.91592
education_binary	1	4.791		2505	1667.9	0.02861 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1						

Religion:

Deviance reduction: 32.7p-value: < 0.05. Religion significantly improves model fit. This indicates that the importance of religion is a strong predictor of happiness.

Tolerance:

Deviance reduction: 5.56 ,p-value < 0.05. Tolerance contributes significantly to explaining happiness, though its effect is moderate compared to other variables.

Authority Respect:

Deviance reduction: 27.32, p-value: < 0.05 . Respect for authority has a statistically significant and meaningful impact on happiness, indicating that attitudes toward authority are important determinants.

Economical Satisfaction:

Deviance reduction: 156.52, p-value: < 0.05 . Economic satisfaction produces the largest reduction in deviance, making it the most influential predictor in the model. This highlights the dominant role of economic conditions in shaping happiness.

Gender:

Deviance reduction: 0.65, p-value: 0.420. Gender does not significantly improve model fit. Happiness levels do not differ meaningfully between males and females after controlling for other factors.

Age:

Deviance reduction: 2.38, p-value: 0.304. Age does not significantly contribute to explaining happiness once other variables are accounted for.

Region:

Deviance reduction: 0.01, p-value: 0.916. There is no evidence that urban–rural residence affects happiness in this model.

Education:

Deviance reduction: 4.79, p-value: 0.029. Education significantly improves model fit. Individuals with higher education levels differ in happiness compared to those with lower or middle education.

- The Analysis of Deviance shows that religion, tolerance, respect for authority, economic satisfaction, and education significantly improve the fit of the logistic regression model. Economic satisfaction emerges as the strongest predictor, followed by religion and respect for authority. In contrast, gender, age, and region do not provide significant explanatory power once other variables are controlled for.

v. Odds Ratios and 95% Confidence Intervals

To facilitate interpretation of the logistic regression results, the estimated coefficients are transformed into odds ratios. Odds ratios provide an intuitive measure of association by indicating how the odds of the outcome change across different values or categories of the

explanatory variables, Confidence intervals are reported to indicate the reliability of the estimates and to assess whether the observed effects are statistically meaningful.

(Intercept)	religionNot Important	toleranceNo	authority_respect.L
3.0991158	0.5423959	0.6920711	1.9066498
authority_respect.Q	economical_satisficationSatisfied	genderF	age.L
0.9257139	4.8940955	1.1622557	1.1265878
age.Q	regionRural	education_binaryHigher	
0.8672255	1.0059587	1.3793075	

	OR	2.5 %	97.5 %
(Intercept)	3.0991158	2.1770800	4.4498773
religionNot Important	0.5423959	0.4144668	0.7084715
toleranceNo	0.6920711	0.5287946	0.9087911
authority_respect.L	1.9066498	1.4796465	2.4491293
authority_respect.Q	0.9257139	0.7423294	1.1513916
economical_satisficationSatisfied	4.8940955	3.7659617	6.4071572
genderF	1.1622557	0.8930401	1.5146828
age.L	1.1265878	0.8840170	1.4353847
age.Q	0.8672255	0.7040527	1.0658030
regionRural	1.0059587	0.6831141	1.5186267
education_binaryHigher	1.3793075	1.0346143	1.8282873

Intercept: OR = 3.10 (95% CI: 2.18 – 4.45)

Represents the baseline odds of being happy for the reference categories. Mainly a statistical reference and not substantively interpreted.

Religion: OR = 0.54 (95% CI: 0.41 – 0.71)

Individuals who consider religion **not important** have **about 46% lower odds** of being happy compared to those who consider it important. The effect is **statistically significant** and substantively meaningful.

Tolerance: OR = 0.69 (95% CI: 0.53 – 0.91)

Respondents who are not tolerant have about 31% lower odds of being happy. This effect is statistically significant, though moderate in size.

Authority Respect: OR = 1.91 (95% CI: 1.48 – 2.45)

Higher respect for authority is associated with nearly double the odds of being happy. This is a strong and statistically significant positive relationship.

Authority Respect (Quadratic Trend): OR = 0.93 (95% CI: 0.74 – 1.15)

The confidence interval includes 1, indicating no significant non-linear effect. The relationship is primarily linear rather than curvilinear.

Economic Satisfaction: OR = 4.89 (95% CI: 3.77 – 6.41)

Economically satisfied individuals have almost five times higher odds of being happy. This is the strongest and most influential predictor in the model.

Gender :OR = 1.16 (95% CI: 0.89 – 1.51)

Females have slightly higher odds of happiness, but the effect is not statistically significant. No meaningful gender difference in happiness after controlling for other factors.

Age: OR = 1.13 (95% CI: 0.88 – 1.44)

The effect of age is not statistically significant. No clear linear age effect on happiness.

Age (Quadratic Component): OR = 0.87 (95% CI: 0.70 – 1.07)

Indicates no significant curvilinear relationship between age and happiness.

Region: OR = 1.01 (95% CI: 0.68 – 1.52)

Living in a rural area does not significantly affect happiness. Regional differences are negligible.

Education: OR = 1.38 (95% CI: 1.03 – 1.83)

Individuals with higher education have about 38% higher odds of being happy. This effect is statistically significant, though moderate in magnitude.

- The odds ratio analysis reveals that economic satisfaction is the strongest determinant of happiness, followed by respect for authority, religion, and education, lack of tolerance and low importance of religion significantly reduce the odds of happiness. In contrast, gender, age, and region do not show statistically significant effects once other variables are controlled for.

- **Multicollinearity Assessment (Variance Inflation Factors)**

Variance Inflation Factors (VIFs) are used to assess **multicollinearity** among the explanatory variables in a regression model. Multicollinearity occurs when predictors are highly correlated, which can inflate standard errors and reduce the reliability of coefficient estimates. As a general rule, VIF values above **5** (or **10** in more conservative guidelines) indicate problematic multicollinearity. For categorical variables with more than one degree of freedom, the **adjusted GVIF^{1/(2·Df)}** is used for interpretation.

	GVIF	Df	GVIF ^{1/(2*Df)}
religion	1.145156	1	1.070120
tolerance	1.046604	1	1.023037
authority_respect	1.120006	2	1.028739
economical_satisfaction	1.023489	1	1.011676
gender	1.111886	1	1.054460
age	1.111330	2	1.026741
region	1.011678	1	1.005822
education_binary	1.038429	1	1.019033

All adjusted VIF values ($GVIF^{1/(2 \cdot Df)}$) range between **1.01 and 1.07**. These values are **very close to 1**, indicating minimal correlation among predictors. No variable approaches commonly accepted thresholds for multicollinearity concern.

- The variance inflation factor analysis indicates that multicollinearity is not a concern in the logistic regression model. All predictors exhibit low VIF values, suggesting that the estimated coefficients are stable and that the effects of individual variables can be interpreted with confidence.

• Model Predictions and Classification Performance

The logistic regression model was used to generate predicted probabilities of happiness. These probabilities were converted into predicted classes using a threshold of 0.5, where values above 0.5 were classified as “happy” (1) and values at or below 0.5 as “not happy” (0). Model performance was evaluated using a confusion matrix and standard classification metrics.

Confusion Matrix:

The confusion matrix shows that the model correctly classifies **2198 happy individuals** (true positives) and only **6 not-happy individuals** (true negatives). However, it misclassifies **309 not-happy individuals** as happy (false positives) and **3 happy individuals** as not happy (false negatives). This indicates that the model is **very effective at**

identifying happy respondents but **fails to correctly identify most not-happy respondents**. The imbalance in predictions reflects the underlying **class imbalance** in the dataset, where the majority of respondents are happy. Overall, while the model demonstrates high sensitivity, it shows **extremely low specificity**, meaning it rarely predicts unhappiness correctly.

	Predicted	
Actual	0	1
0	6	309
1	3	2198

Performance Measures:

- **Accuracy : 0.876**

Accuracy: 0.876

The model correctly classifies **87.6%** of all observations. This high value is influenced by the large proportion of happy individuals in the sample.

- **Sensitivity : 0.999 :**

Sensitivity: 0.999

The model correctly identifies **99.9% of happy individuals**. This indicates excellent performance in detecting positive outcomes (happiness).

- **Specificity = 0.019:**

Specificity: 0.019

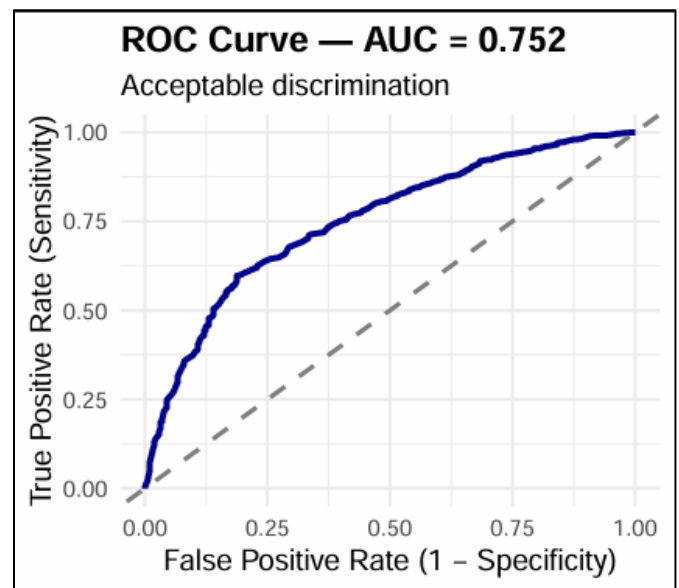
The model correctly identifies only **1.9% of not-happy individuals**. This indicates very poor performance in detecting negative outcomes.

- Using a 0.5 classification threshold, the logistic regression model achieves high overall accuracy and near-perfect sensitivity but very low specificity. While the model is effective at identifying happy individuals, it performs poorly in distinguishing those who are not happy. This suggests that alternative thresholds or additional performance measures may be required for a more balanced classification.

- **ROC Curve and AUC**

The Receiver Operating Characteristic (ROC) curve illustrates the model's ability to discriminate between happy and not-happy respondents across all possible classification thresholds. The curve plots the True Positive Rate (Sensitivity) against the False Positive Rate (1 – Specificity). **AUC (Area Under the Curve) = 0.752**

This indicates that the model has **acceptable discrimination**. In practical terms, there is a **75.2% chance** that the model will correctly distinguish a randomly chosen happy individual from a randomly chosen not-happy individual.



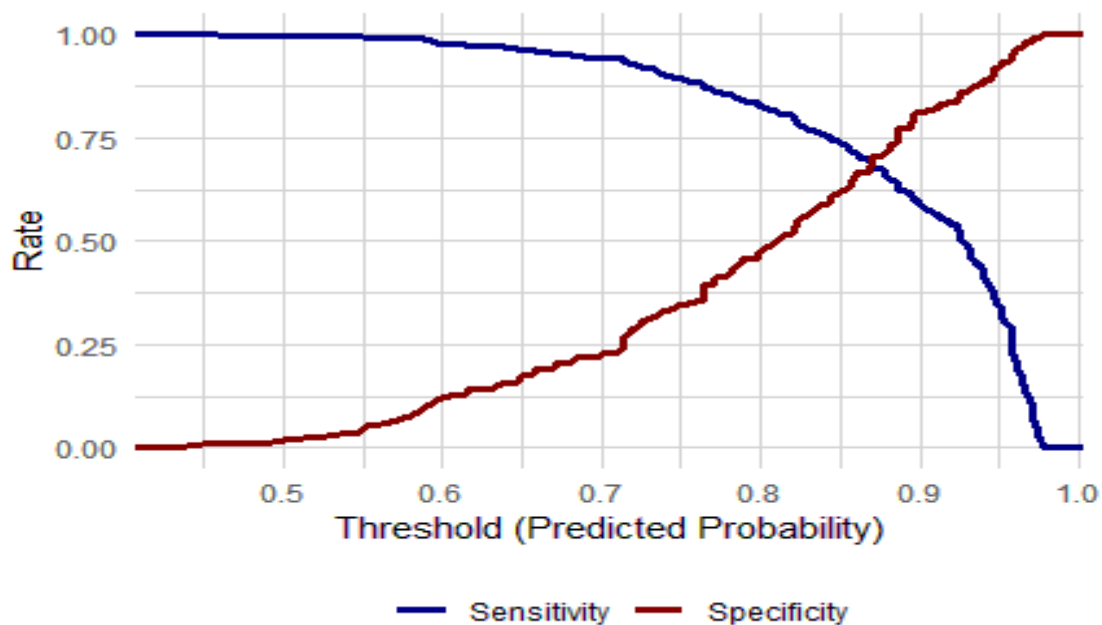
ROC Curve Shape

The curve rises above the diagonal reference line, indicating **better-than-random classification**. The curve does not reach the top-left corner perfectly, reflecting **imperfect sensitivity and specificity trade-off**.

While the model is reasonably good at distinguishing happy from not-happy respondents, there is room for improvement. The ROC curve confirms earlier findings: the model favors predicting happiness (high sensitivity), but specificity remains low.

- The ROC curve and AUC value demonstrate that the logistic regression model achieves **acceptable predictive performance**. Although the model effectively identifies happy individuals, it shows limited ability to correctly classify not-happy individuals, consistent with the observed low specificity.

Sensitivity and Specificity across Classification Thresholds:



The figure illustrates how sensitivity and specificity vary across different classification thresholds for the logistic regression model. Sensitivity represents the model's ability to correctly identify individuals who are happy (true positives), while specificity reflects the ability to correctly classify individuals who are not happy (true negatives). As shown in the figure, sensitivity remains very high at low classification thresholds, indicating that the model successfully captures the majority of happy individuals when a lenient cutoff is applied. However, this comes at the cost of low specificity, as many non-happy individuals

are incorrectly classified as happy. As the threshold increases, sensitivity gradually declines, while specificity steadily improves. This trade-off reflects the inherent balance between correctly identifying positive cases and avoiding false positives. At higher threshold values, specificity approaches one, meaning that the model becomes very conservative in predicting happiness and correctly classifies most non-happy individuals. However, this increased specificity is accompanied by a sharp decline in sensitivity, implying that a substantial proportion of truly happy individuals are misclassified as not happy. The intersection region where sensitivity and specificity converge highlights a range of thresholds where the trade-off between the two measures becomes more balanced.

Optimal Threshold Selection Using Youden's Index:

To determine an appropriate classification threshold, Youden's index was employed as an objective criterion that jointly maximizes sensitivity and specificity. This approach is particularly suitable in contexts where the default threshold of 0.5 may lead to suboptimal classification performance due to class imbalance, as is the case in the present dataset. The optimal threshold identified by Youden's index is 0.899, corresponding to a sensitivity of 0.597 and a specificity of 0.813. This threshold represents the point at which the difference between the true positive rate and the false positive rate is maximized, yielding the best overall discriminatory power between happy and not-happy individuals. Out of the individuals who are not happy, 256 are correctly classified, while 59 are misclassified as happy. Among happy individuals, 1,315 are correctly identified, whereas 886 are incorrectly classified as not happy. The relatively large number of false negatives reflects the conservative nature of the selected threshold, which prioritizes correct identification of non-happy individuals. Using the optimal threshold, the model achieves an overall classification accuracy of 62.4%, with a sensitivity of 59.7% and a specificity of 81.3%. Compared to lower thresholds, this specification substantially improves specificity at the cost of reduced sensitivity. This trade-off is consistent with the pattern observed in the sensitivity–specificity curves and aligns with the objective of minimizing false positive classifications of happiness.

MODEL SELECTION: STEPWISE REGRESSION:

Start: AIC=1689.88

happiness_binary ~ religion + tolerance + authority_respect +
economical_satisfaction + gender + age + region + education_binary

	Df	Deviance	AIC
- region	1	1667.9	1687.9
- age	2	1670.6	1688.6
- gender	1	1669.1	1689.1
<none>		1667.9	1689.9
- education_binary	1	1672.7	1692.7
- tolerance	1	1674.8	1694.8
- religion	1	1688.1	1708.1
- authority_respect	2	1692.6	1710.6
- economical_satisfaction	1	1821.0	1841.0

This output reports the first step of the backward stepwise regression procedure based on the Akaike Information Criterion (AIC). Starting from the full logistic regression model, each explanatory variable is evaluated for potential removal. The results show that excluding region yields the largest reduction in AIC at this stage, indicating that region contributes the least to model fit once other covariates are included. In contrast, removing economic satisfaction or respect for authority substantially increases the AIC, highlighting their strong explanatory power. This initial step provides evidence that not all demographic variables contribute equally to explaining happiness in the multivariate context.

Step: AIC=1687.88

happiness_binary ~ religion + tolerance + authority_respect +
economical_satisfaction + gender + age + education_binary

	Df	Deviance	AIC
- age	2	1670.6	1686.6
- gender	1	1669.1	1687.1
<none>		1667.9	1687.9
- education_binary	1	1672.7	1690.7
- tolerance	1	1674.8	1692.8
- religion	1	1688.1	1706.1
- authority_respect	2	1692.6	1708.6
- economical_satisfaction	1	1821.0	1839.0

This output presents the second step of the backward elimination process after removing the region variable. At this stage, age emerges as the next candidate for exclusion, as its removal leads to a further decrease in AIC. This suggests that the effect of age on happiness observed in earlier analyses is largely absorbed by other socio-economic and attitudinal variables in the model. The continued stability of key predictors, such as economic

satisfaction and respect for authority, confirms their robustness across different model specifications.

Step: AIC=1686.62

happiness_binary ~ religion + tolerance + authority_respect +
economical_satisfaction + gender + education_binary

	Df	Deviance	AIC
- gender	1	1671.5	1685.5
<none>		1670.6	1686.6
- education_binary	1	1675.1	1689.1
- tolerance	1	1677.7	1691.7
- religion	1	1691.7	1705.7
- authority_respect	2	1696.5	1708.5
- economical_satisfaction	1	1824.5	1838.5

This output presents that the model still includes gender alongside religion, tolerance, respect for authority, economic satisfaction, and education. The table shows that removing gender leads to a further reduction in the AIC from 1686.62 to 1685.51, indicating that gender does not provide additional explanatory power once attitudinal, economic, and educational factors are controlled for. In contrast, removing any of the remaining predictors results in a notable increase in the AIC, particularly for economic satisfaction and respect for authority, underscoring their central role in explaining happiness.

Step: AIC=1685.51

happiness_binary ~ religion + tolerance + authority_respect +
economical_satisfaction + education_binary

	Df	Deviance	AIC
<none>		1671.5	1685.5
- education_binary	1	1675.7	1687.7
- tolerance	1	1679.2	1691.2
- religion	1	1693.7	1705.7
- authority_respect	2	1697.8	1707.8
- economical_satisfaction	1	1824.6	1836.6

This output reports the final model selected by the backward stepwise regression procedure based on the Akaike Information Criterion. No further variable removal leads to an improvement in the AIC, confirming that this specification represents the optimal balance between model fit and parsimony. The final model excludes demographic variables such as age, gender, and region, while retaining key attitudinal, economic, and educational predictors. This result suggests that the influence of demographic characteristics on happiness operates largely through these retained variables rather than exerting an independent effect. So, the final model will be:

$$\log \left(\frac{P(Happiness_i = 1)}{1 - P(Happiness_i = 1)} \right) = \beta_0 + \beta_1 Religion_i + \beta_2 Tolerance_i + \beta_3 AuthorityRespect_i^{(L)} + \beta_4 AuthorityRespect_i^{(Q)} + \beta_5 EconomicSatisfaction_i + \beta_6 Education_i$$

Call:

```
glm(formula = happiness_binary ~ religion + tolerance + authority_respect +
    economical_satisfaction + education_binary, family = binomial(link = "logit"),
    data = df_model)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.2539	0.1577	7.951	1.85e-15 ***
religionNot Important	-0.6331	0.1350	-4.690	2.73e-06 ***
toleranceNo	-0.3834	0.1371	-2.796	0.00518 **
authority_respect.L	0.6621	0.1277	5.183	2.19e-07 ***
authority_respect.Q	-0.0792	0.1113	-0.711	0.47684
economical_satisfactionSatisfied	1.5749	0.1343	11.727	< 2e-16 ***
education_binaryHigher	0.2988	0.1441	2.074	0.03811 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 1897.9 on 2515 degrees of freedom
 Residual deviance: 1671.5 on 2509 degrees of freedom
 AIC: 1685.5

This output presents the coefficient estimates of the final stepwise logistic regression model. So, we can get estimated model as:

$$\log \left(\frac{P(Happiness_i = 1)}{1 - P(Happiness_i = 1)} \right) = 1.2539 - 0.6331 Religion_i - 0.3834 Tolerance_i + 0.6621 AuthorityRespect_i^{(L)} - 0.0792 AuthorityRespect_i^{(Q)} + 1.5749 EconomicSatisfaction_i + 0.2988 Education_i$$

Economic satisfaction emerges as the strongest predictor of happiness, with a large positive and highly statistically significant coefficient, indicating that economically satisfied individuals are substantially more likely to report being happy. Respect for authority also shows a strong positive association, particularly along the linear trend, suggesting that higher levels of respect are linked to greater happiness. Religion and tolerance remain significant, with negative coefficients for lower importance of religion and lack of tolerance, respectively, implying reduced likelihood of happiness among these groups. Higher education retains a positive and statistically significant effect, although its magnitude is comparatively smaller than that of economic and attitudinal factors.

Full Model AIC: 1689.876

Stepwise Model AIC: 1685.51

The full model has an AIC of 1689.88, while the stepwise model achieves a lower AIC of 1685.51. This reduction of 4.37 points indicates an improvement in model parsimony without sacrificing explanatory power. The comparison provides empirical justification for favoring the stepwise-selected model, as it achieves a better balance between fit and complexity with fewer parameters. The result confirms the robustness of the retained predictors religion, tolerance, respect for authority, economic satisfaction, and education and supports the use of the stepwise model as a valid alternative to the full specification.

COMPARING MODELS WITH ORIGINAL VS RECODED EDUCATION:

The comparison between the full model using the original education variable and the recoded education variable shows a slight improvement in model fit with recoding. Specifically, the full model with the original education variable has an AIC of 1690.29, whereas the model with the recoded education variable achieves a slightly lower AIC of 1689.88. This indicates a modest gain in parsimony without compromising model performance. The classification performance at the default 0.5 threshold is summarized in the confusion matrices below.

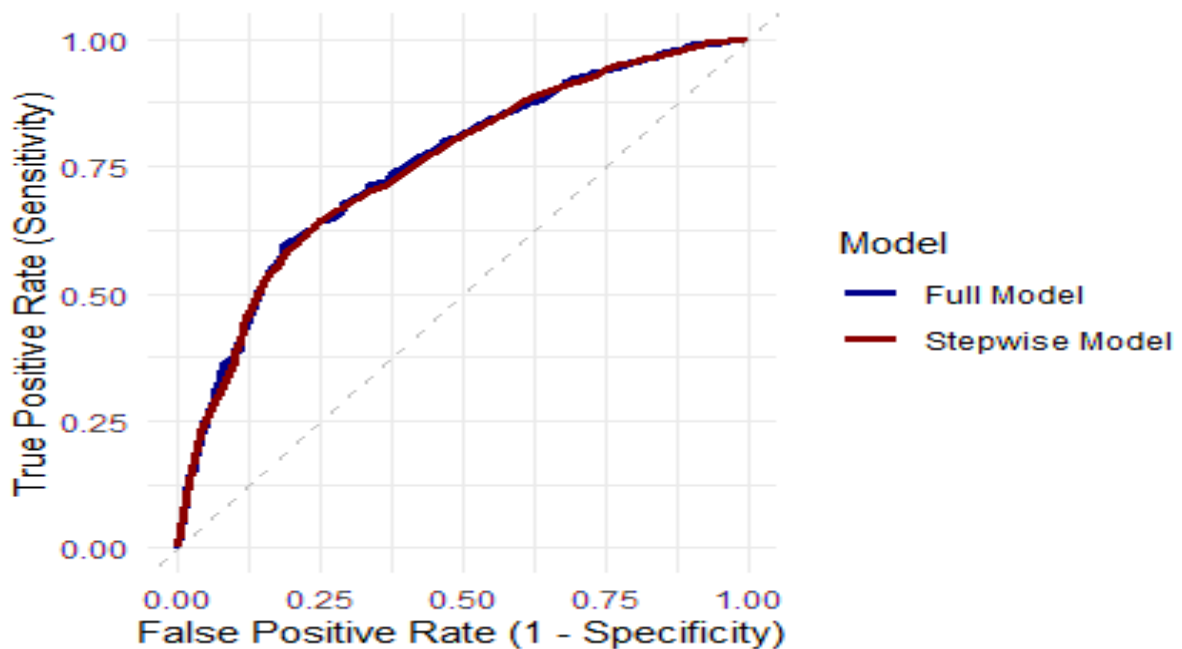
- Full Model Confusion Matrix

Actual \ Predicted	0	1
0	6	309
1	3	2198

- Stepwise Model Confusion Matrix

Actual \ Predicted	0	1
0	3	312
1	1	2200

The overall classification metrics show that both models have high accuracy (full model: 87.60%, stepwise: 87.56%) and extremely high sensitivity (full model: 99.86%, stepwise: 99.95%), but very low specificity (full model: 1.90%, stepwise: 0.95%), reflecting that the models are more effective at correctly predicting “happy” respondents.



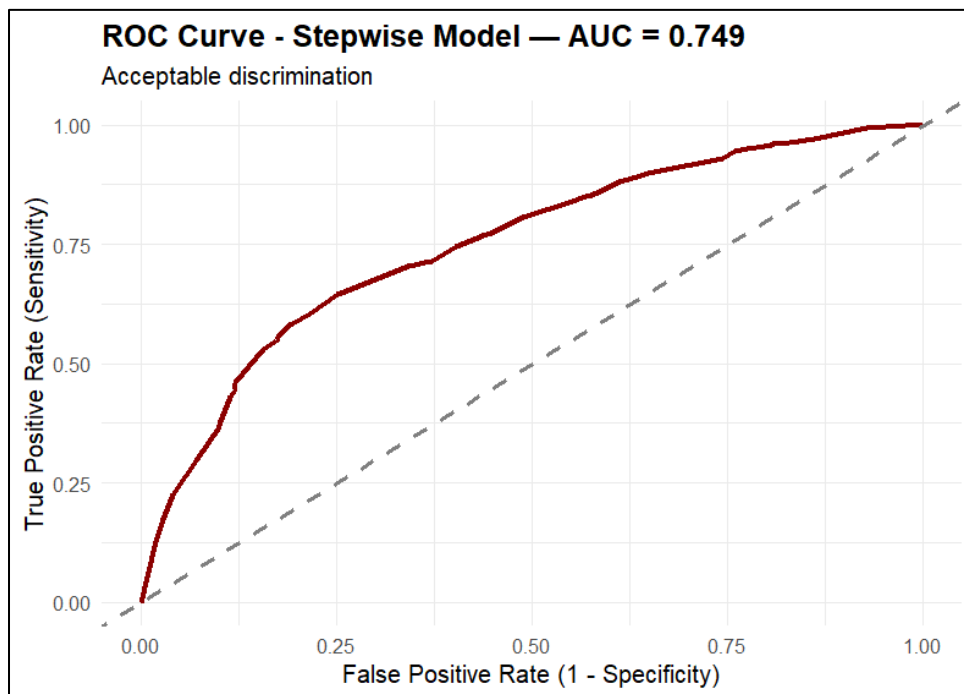
The ROC curve comparison Full vs Stepwise Model, demonstrates comparable discriminative ability with the full model AUC at 0.752 and the stepwise model AUC at 0.749, indicating that the reduced stepwise model maintains predictive power while simplifying the model.

Model Performance Analysis: Stepwise Model

1. ROC Curve:

The Stepwise Model was evaluated using ROC curve analysis to determine its ability to differentiate between classes.

- **Overall Accuracy:** The model achieved an Area Under the Curve (AUC) of **0.749**. In statistical practice, this is classified as "**Acceptable Discrimination**," meaning that in a randomly selected pair of positive and negative cases, the model will correctly rank the positive case higher approximately 75% of the time.
- **Curve Characteristics:** The ROC curve shows a steep initial rise, particularly in the lower False Positive Rate (FPR) region. This indicates that the model is highly effective at identifying true positives while maintaining high specificity (low "false alarms").
- **Comparison to Baseline:** The curve maintains a significant distance from the diagonal reference line (random chance), confirming that the selected predictors provide meaningful classification power.



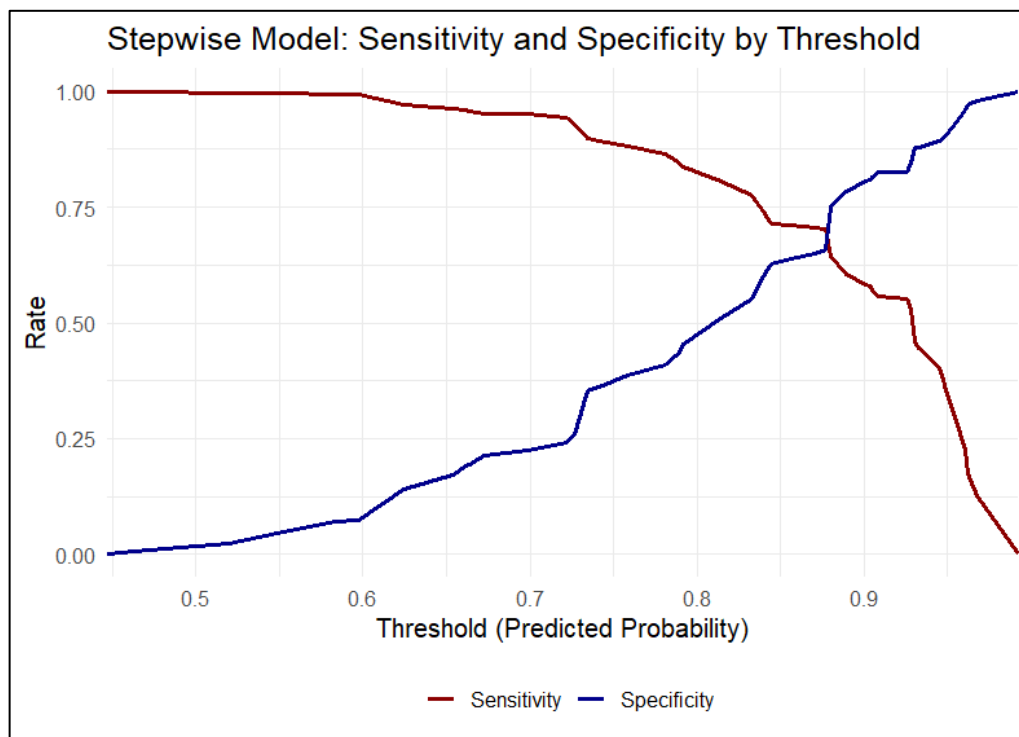
The Stepwise Model demonstrates acceptable discriminatory ability with an AUC = 0.749.

While it does not reach the 'Excellent' threshold (typically 0.80), the model performs well at higher specificity levels and provides a statistically significant improvement over random classification.

2. Sensitivity and Specificity by Threshold:

This analysis reveals the classic inverse relationship between **Sensitivity** (True Positive Rate) and **Specificity** (True Negative Rate). As the classification threshold changes, the model's ability to identify positives and negatives shifts in opposite directions:

- **Low Thresholds (0.5–0.6):** The model is highly "inclusive," achieving near-perfect Sensitivity (100%). However, this comes at the cost of very low Specificity, as the model becomes "greedy" and misclassifies many negatives as positives (high false-positive rate).
- **High Thresholds (>0.95):** The model becomes highly "selective," where Specificity approaches 100%. At this level, the model rarely misidentifies a negative case, but its Sensitivity drops dramatically, failing to catch many true positive cases.
- **The Optimal Balance:** The curves intersect at a threshold of approximately **0.87–0.88**. At this "crossover point," both metrics are balanced at roughly



70%, representing the point of maximum combined accuracy for general classification.

Strategic Selection: The final threshold should be selected based on the specific cost of error:

- **Prioritize Sensitivity** when missing a case (false negative) is dangerous, such as in early disease screening.
- **Prioritize Specificity** when a false alarm (false positive) has severe consequences, such as invasive confirmatory surgery or legal judgment.

The gradual slopes in the **0.7–0.9 range** provide a flexible where the threshold can be fine-tuned to meet these specific clinical or practical requirements without a total collapse in performance.

3. Optimal Threshold

- **Threshold: 0.8798777**

This specific threshold represents the exact bin where the model's predictive power for both classes is most symmetrical.

- **Confusion Matrix**

The matrix shows the following raw counts:

- **True Negatives:** 236
- **False Positives:** 79
- **False Negatives:** 781
- **True Positives:** 1420

		Predicted	
Actual	0	1	
	0	236	79
	1	781	1420

Accuracy (0.658): The model correctly predicts the outcome in approximately 65.8% of all cases.

Sensitivity (0.645): The model successfully identifies 64.5% of actual positive instances.

Specificity (0.749): The model successfully identifies 74.9% of actual negative instances.

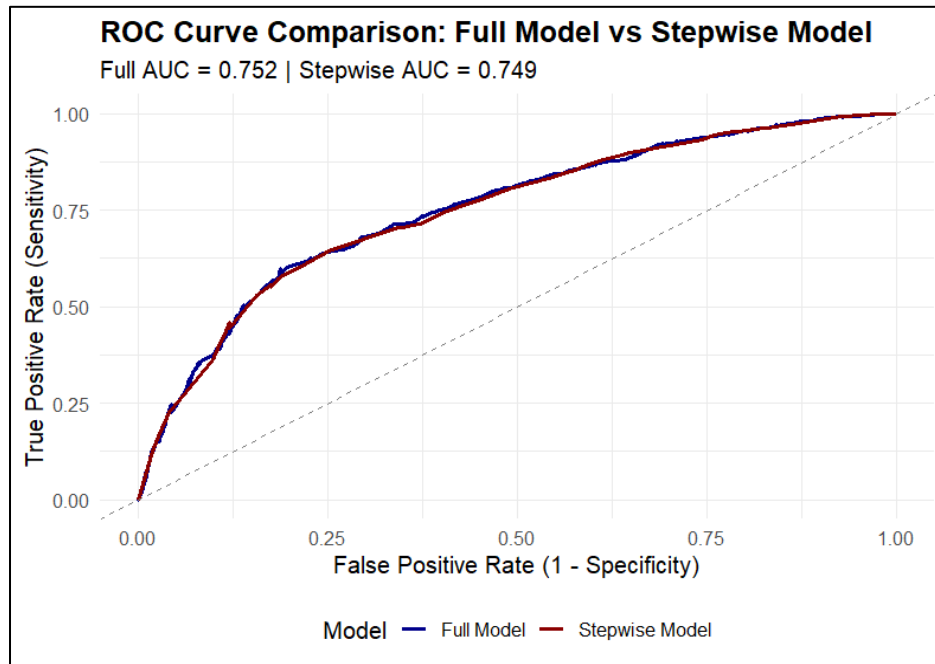
The threshold of 0.88 serves as a strategic balance point on the threshold plot.

- **Balanced Trade-off:** At this stage, the model is slightly more effective at identifying negative cases (Specificity) than positive cases (Sensitivity).
- **Alignment with Visual Data:** These metrics align with the "crossing point" in the visualization, where the red Sensitivity line and blue Specificity line converge.

- **Practical Application:** With a Sensitivity of 0.645 and Specificity of 0.749, this model is moderately conservative. It is leaning toward ensuring that when it predicts a negative, it is likely correct, at the expense of missing about 35.5% of positive cases.

Compare Full Model VS Stepwise Model:

The visual and numerical data indicate that both models demonstrate good predictive performance, significantly exceeding the baseline of a random classifier (the diagonal dashed line).



Key Insights

- **AUC Performance:** The Full Model achieves an Area Under the Curve (AUC) of 0.752, while the Stepwise Model follows closely with an AUC of 0.749.
- **Marginal Difference:** There is a negligible difference of only 0.003 in the AUC between the two approaches. The ROC curves are nearly indistinguishable across all thresholds for False Positive and True Positive rates.
- **Model Parsimony:** Given the high degree of overlap, the Stepwise Model is likely the preferred choice. It achieves nearly identical predictive power while potentially utilizing fewer predictors, which enhances model interpretability and reduces the risk of overfitting.

According to ROC curve both models show comparable discriminative ability. The Stepwise Model is recommended for deployment due to its efficiency and simplicity, as the inclusion of additional variables in the Full Model yields no significant gain in sensitivity or specificity.

Choose STEPWISE MODEL:

- Similar performance (AUC difference < 0.02)
- Lower AIC (better parsimony)
- Fewer predictors (simpler model).

General conclusion:

The statistical analysis conducted in this project provides clear empirical evidence on the key factors associated with individual happiness. Using binary logistic regression, the study demonstrates that happiness is not randomly distributed across individuals, but is systematically related to a specific set of economic and attitudinal determinants.

The results consistently identify **economic satisfaction** as the most influential predictor of happiness. Individuals who are economically satisfied exhibit substantially higher odds of reporting happiness compared to dissatisfied individuals, making economic well-being the dominant explanatory factor in the model. This effect remains strong and statistically significant across all model specifications, diagnostic tests, and model selection procedures, confirming its robustness.

Beyond economic factors, **attitudinal variables** play an important and independent role. Higher respect for authority is associated with significantly increased odds of happiness, indicating a positive link between institutional attitudes and subjective well-being. Similarly, individuals who consider religion important and those who exhibit tolerant social attitudes are more likely to report happiness, even after controlling for other covariates. These findings suggest that value-based orientations contribute meaningfully to happiness alongside material conditions.

Educational attainment also shows a positive and statistically significant association with happiness. After recoding to address functional form issues, individuals with higher education display higher odds of happiness, indicating that education contributes to well-being beyond its indirect economic effects.

In contrast, **demographic variables** such as gender, age, and region of residence do not have statistically significant independent effects once economic and attitudinal factors are accounted for. This implies that apparent demographic differences in happiness operate largely through underlying socio-economic and value-related mechanisms rather than demographic characteristics themselves.

From a modeling perspective, the stepwise selection procedure confirms that a **parsimonious specification** retaining economic satisfaction, religion, tolerance, respect for authority, and education achieves nearly identical predictive performance to the full model while improving interpretability. Although the model demonstrates acceptable discriminatory power, classification results reveal a strong sensitivity–specificity trade-off driven by class imbalance, emphasizing that predictive accuracy alone should not be used to assess model quality.

Overall, the statistical findings indicate that happiness is primarily shaped by economic perceptions and individual value systems, with limited direct influence from basic demographic traits. The results highlight the importance of multivariate statistical modeling in uncovering the underlying structure of happiness and provide a rigorous empirical foundation for understanding its key determinants.

Data Source Reference:

World Values Survey Association. (2022). World Values Survey: Wave 7 (2017–2020).

Official aggregate data file.

Madrid, Spain: JD Systems Institute.

Available at: <https://www.worldvaluessurvey.org>

Appendix:

```

458 # source: test_care_cmm2/
459 #####
460 #model
461 #####
462 # =====
463 # PART 2: PREDICTIVE MODELING FOR USA HAPPINESS DATA
464 # =====
465
466 # Load necessary packages
467 library(tidyverse)
468 library(broom)
469 library(car)
470 library(ResourceSelection)
471 library(ROCR)
472 library(caret)
473 library(pscl)
474 library(nnet)
475 library(MASS)
476 library(pROC)
477
478 # Assuming df_final from Part 1 is already loaded
479 # If not, run the data cleaning code from Part 1 first
480
481 # =====
482 # 1. DATA PREPARATION FOR MODELING
483 # =====
484
485 # Create a binary numeric version of happiness for modeling
486 df_model <- df_final %>%
487   mutate(
488     happiness_binary = ifelse(happiness == "HAPPY", 1, 0)
489   )
490
491 # Check the distribution
492 table(df_model$happiness_binary)
493 prop.table(table(df_model$happiness_binary))
494
495 # Verify all variables
496 str(df_model)
497 summary(df_model)
498
499 # =====
500 # 2. S-SHAPE ANALYSIS: CHECK LOGISTIC RELATIONSHIP
501 # =====
502
503 cat("\n=== Checking S-Shape Pattern for Logistic Regression ===\n")
504
505 # For ordinal variables, check if proportions follow S-shape pattern
506
507 # 2.1: Age groups
508 prop_age <- df_model %>%
509   group_by(age) %>%
510   summarize(
511     n_total = n(),
512     n_happy = sum(happiness_binary == 1),
513     prop_happy = n_happy / n_total
514   )
515
516 print("Proportions by Age:")
517 print(prop_age)
518
519 ggplot(prop_age, aes(x = age, y = prop_happy, group = 1)) +
520   geom_point(size = 4, color = "darkblue") +
521   geom_line(color = "darkblue", size = 1.2) +
522   labs(
523     title = "S-Shape Check: Proportion of Happy People by Age",
524     x = "Age Group",
525     y = "Proportion Happy"
526   ) +
527   theme_minimal() +
528   theme(axis.text.x = element_text(angle = 45, hjust = 1))
529
530 # 2.2: Education levels (BEFORE recoding)
531 prop_education <- df_model %>%
532   group_by(education) %>%
533   summarize(
534     n_total = n(),
535     n_happy = sum(happiness_binary == 1),
536     prop_happy = n_happy / n_total
537   )
538
539 print("Proportions by Education (Original):")
540 print(prop_education)
541
542 ggplot(prop_education, aes(x = education, y = prop_happy, group = 1)) +
543   geom_point(size = 4, color = "darkgreen") +

```



```

543 geom_point(size = 4, color = "darkgreen") +
544 geom_line(color = "darkgreen", size = 1.2) +
545 labs(
546   title = "S-Shape Check: Proportion of Happy People by Education (Original)",
547   x = "Education Level",
548   y = "Proportion Happy"
549 ) +
550 theme_minimal() +
551 theme(axis.text.x = element_text(angle = 45, hjust = 1))
552
553 cat("\n*** EDUCATION SHOWS U-SHAPE - RECODING NEEDED ***\n")
554
555 # 2.3: Authority respect
556 prop_authority <- df_model %>%
557   group_by(authority_respect) %>%
558   summarize(
559     n_total = n(),
560     n_happy = sum(happiness_binary == 1),
561     prop_happy = n_happy / n_total
562   )
563
564 print("Proportions by Authority Respect:")
565 print(prop_authority)
566
567 ggplot(prop_authority, aes(x = authority_respect, y = prop_happy, group = 1)) +
568   geom_point(size = 4, color = "darkorange") +
569   geom_line(color = "darkorange", size = 1.2) +
570   labs(
571     title = "S-Shape Check: Proportion of Happy People by Authority Respect",
572     x = "Authority Respect",
573     y = "Proportion Happy"
574   ) +
575   theme_minimal() +
576   theme(axis.text.x = element_text(angle = 45, hjust = 1))
577
578
579 # 2.5: Binary variables
580 prop_religion <- df_model %>%
581   group_by(religion) %>%
582   summarize(
583     n_total = n(),
584     n_happy = sum(happiness_binary == 1),
585     prop_happy = n_happy / n_total
586   )
587
588
589 print("Proportions by Religion:")
590 print(prop_religion)
591
592 prop_gender <- df_model %>%
593   group_by(gender) %>%
594   summarize(
595     n_total = n(),
596     n_happy = sum(happiness_binary == 1),
597     prop_happy = n_happy / n_total
598   )
599
600 print("Proportions by Gender:")
601 print(prop_gender)
602
603 prop_region <- df_model %>%
604   group_by(region) %>%
605   summarize(
606     n_total = n(),
607     n_happy = sum(happiness_binary == 1),
608     prop_happy = n_happy / n_total
609   )
610
611 print("Proportions by Region:")
612 print(prop_region)
613
614 prop_tolerance <- df_model %>%
615   group_by(tolerance) %>%
616   summarize(
617     n_total = n(),
618     n_happy = sum(happiness_binary == 1),
619     prop_happy = n_happy / n_total
620   )
621
622 print("Proportions by Tolerance:")
623 print(prop_tolerance)
624
625 prop_economic <- df_model %>%
626   group_by(economical_satisfaction) %>%
627   summarize(
628     n_total = n(),

```

```

627   summarize(
628     n_total = n(),
629     n_happy = sum(happiness_binary == 1),
630     prop_happy = n_happy / n_total
631   )
632
633   print("Proportions by economical_satisfaction:")
634   print(prop_economic)
635
636
637
638
639 - # =====
640 # 3. RECODE EDUCATION TO AVOID U-SHAPE
641 - # =====
642
643   cat("\n=== RECODING EDUCATION VARIABLE ===\n")
644
645   # Recode education: Middle vs Lower/Higher
646   df_model <- df_model %>%
647     mutate(education_binary = ifelse(education == c("Lower", "Middle"), "Lower/Middle", "Higher"))
648
649   # Convert to factor
650   df_model$education_binary <- factor(df_model$education_binary,
651     levels = c("Lower/Middle", "Higher"))
652
653   # Check the recoded variable
654   table(df_model$education_binary)
655
656   # Check proportions after recoding
657   prop_education_recoded <- df_model %>%
658     group_by(education_binary) %>%
659     summarize(
660       n_total = n(),
661       n_happy = sum(happiness_binary == 1),
662       prop_happy = n_happy / n_total
663     )
664
665   print("Proportions by Education (Recoded):")
666   print(prop_education_recoded)
667
668   ggplot(prop_education_recoded, aes(x = education_binary, y = prop_happy, group = 1)) +
669     geom_point(size = 4, color = "purple") +
670     geom_line(color = "purple", size = 1.2) +
671     labs(
672       title = "Proportion of Happy People by Education (Recoded)",
673       x = "Education Level",
674       y = "Proportion Happy"
675     ) +
676     theme_minimal() +
677     theme(axis.text.x = element_text(angle = 45, hjust = 1))
678
679 - # =====
680 # 4. BINARY LOGISTIC REGRESSION MODEL
681 - # =====
682
683   cat("\n=== BINARY LOGISTIC REGRESSION MODEL ===\n")
684
685   # Fit the full model with all predictors (using recoded education)
686   logit_model_full <- glm(
687     happiness_binary ~ religion + tolerance + authority_respect +
688     economical_satisfaction + gender + age + region + education_binary,
689     data = df_model,
690     family = binomial(link = "logit")
691   )
692
693   # Model summary
694   summary(logit_model_full)
695
696 - # =====
697 # 5. MODEL DIAGNOSTICS AND GOODNESS OF FIT
698 - # =====
699
700   # Pseudo R-squared values
701   cat("\n=== Pseudo R-squared values ===\n")
702   pseudo_r2 <- pr2(logit_model_full)
703   print(pseudo_r2)
704
705   # Hosmer-Lemeshow test
706   cat("\n=== Hosmer-Lemeshow Goodness of Fit Test ===\n")
707   hl_test <- hoslem.test(df_model$happiness_binary, fitted(logit_model_full))
708   print(hl_test)
709
710   # Likelihood Ratio Test
711   cat("\n=== Likelihood Ratio Test ===\n")
712   deviance_diff <- with(logit_model_full, null.deviance - deviance)
713   df_diff <- with(logit_model_full, df.null - df.residual)
714   p_value_lrt <- with(logit_model_full,
715     1 - pchisq(null.deviance - deviance,
716       df.null - df.residual))
717
718   cat("Deviance difference:", deviance_diff, "\n")
719   cat("DF difference:", df_diff, "\n")
720   cat("P-value:", p_value_lrt, "\n")
721
722   # Analysis of Deviance
723   cat("\n=== Analysis of Deviance ===\n")

```

```

726 # =====
727 # 6. ODDS RATIOS AND CONFIDENCE INTERVALS
728 # =====
729
730 cat("\n=== Odds Ratios ===\n")
731 odds_ratios <- exp(coef(logit_model_full))
732 print(odds_ratios)
733
734 cat("\n=== Odds Ratios with 95% Confidence Intervals ===\n")
735 or_ci <- exp(cbind(OR = coef(logit_model_full),
736                   confint(logit_model_full)))
737 print(or_ci)
738
739 # =====
740 # 7. MULTICOLLINEARITY CHECK
741 # =====
742
743 cat("\n=== Variance Inflation Factors (VIF) ===\n")
744 vif_values <- vif(logit_model_full)
745 print(vif_values)
746
747 # =====
748 # 8. MODEL PREDICTIONS AND PERFORMANCE
749 # =====
750
751 # Generate predictions
752 df_model$pred_prob <- predict(logit_model_full, type = "response")
753 df_model$pred_class <- ifelse(df_model$pred_prob > 0.5, 1, 0)
754
755 # Confusion matrix (default threshold = 0.5)
756 cat("\n=== Confusion Matrix (Threshold = 0.5) ===\n")
757 conf_matrix <- table(Actual = df_model$happiness_binary,
758                     Predicted = df_model$pred_class)
759 print(conf_matrix)
760
761 # Calculate accuracy, sensitivity, specificity
762 accuracy <- sum(diag(conf_matrix)) / sum(conf_matrix)
763 sensitivity <- conf_matrix[2,2] / sum(conf_matrix[2,])
764 specificity <- conf_matrix[1,1] / sum(conf_matrix[1,])
765
766 cat("\nAccuracy:", round(accuracy, 3), "\n")
767 cat("Sensitivity:", round(sensitivity, 3), "\n")
768 cat("Specificity:", round(specificity, 3), "\n")
769
770 # =====
771 # 9. ROC CURVE AND AUC
772 # =====
773
774 # Calculate AUC
775 auc_value <- round(as.numeric(auc(roc_obj)), 3)
776 cat("\n=== Area Under the Curve (AUC) ===\n")
777 cat("AUC:", auc_value, "\n")
778
779 # Plot ROC curve
780 ggroc(roc_obj, legacy.axes = TRUE, size = 1.4, colour = "darkblue") +
781   geom_abline(intercept = 0, slope = 1,
782              linetype = "dashed", colour = "gray50", size = 1) +
783   labs(
784     title = paste("ROC Curve - AUC =", auc_value),
785     subtitle = ifelse(auc_value >= 0.8, "Excellent discrimination",
786                      ifelse(auc_value >= 0.7, "Acceptable discrimination",
787                             "Poor discrimination")),
788     x = "False Positive Rate (1 - Specificity)",
789     y = "True Positive Rate (Sensitivity)"
790   ) +
791   theme_minimal(base_size = 14) +
792   theme(plot.title = element_text(face = "bold"))
793
794 # =====
795 # 10. SENSITIVITY AND SPECIFICITY BY THRESHOLD
796 # =====
797
798 # Extract sensitivity and specificity at different thresholds
799 roc_data <- coords(roc_obj, "all",
800                   ret = c("threshold", "sensitivity", "specificity"),
801                   transpose = FALSE) %>%
802   as.data.frame() %>%
803   pivot_longer(cols = c(sensitivity, specificity),
804               names_to = "metric",
805               values_to = "value")
806
807 # Plot sensitivity and specificity
808 ggplot(roc_data, aes(x = threshold, y = value, color = metric)) +
809   geom_line(size = 1.2) +
810   scale_color_manual(
811     values = c("sensitivity" = "darkblue", "specificity" = "darkred"),
812     labels = c("Sensitivity", "Specificity")
813   ) +
814   labs(
815     title = "Sensitivity and Specificity by Classification Threshold",
816     x = "Threshold (Predicted Probability)",
817     y = "Rate",
818     color = ""
819   ) +
820   theme_minimal() +
821   theme(legend.position = "bottom")
822
823 # =====
824
825
826 # =====

```

```

826 # =====
827 # 11. OPTIMAL THRESHOLD SELECTION
828 # =====
829
830 # Find optimal threshold using Youden's index
831 optimal_coords <- coords(roc_obj, "best",
832   ret = c("threshold", "sensitivity", "specificity"),
833   best.method = "youden")
834 cat("\n=== Optimal Threshold (Youden's Index) ===\n")
835 print(optimal_coords)
836
837 # Apply optimal threshold
838 optimal_threshold <- optimal_coords$threshold
839 df_model$pred_class_optimal <- ifelse(df_model$pred_prob > optimal_threshold, 1, 0)
840
841 # Confusion matrix with optimal threshold
842 cat("\n=== Confusion Matrix (Optimal Threshold) ===\n")
843 conf_matrix_opt <- table(Actual = df_model$happiness_binary,
844   Predicted = df_model$pred_class_optimal)
845 print(conf_matrix_opt)
846
847 accuracy <- sum(diag(conf_matrix_opt)) / sum(conf_matrix_opt)
848 sensitivity <- conf_matrix_opt[2,2] / sum(conf_matrix_opt[2,])
849 specificity <- conf_matrix_opt[1,1] / sum(conf_matrix_opt[1,])
850 cat("\nAccuracy:", round(accuracy, 3), "\n")
851 cat("Sensitivity:", round(sensitivity, 3), "\n")
852 cat("Specificity:", round(specificity, 3), "\n")
853
854 # =====
855 # 12. MODEL SELECTION: STEPWISE REGRESSION
856 # =====
857
858 cat("\n=== Stepwise Model Selection (AIC) ===\n")
859
860 # Backward elimination
861 logit_model_step <- step(logit_model_full, direction = "backward",
862   trace = 1)
863
864 # Summary of the selected model
865 summary(logit_model_step)
866
867 # Compare AIC values
868 cat("\nFull Model AIC:", AIC(logit_model_full), "\n")
869 cat("Stepwise Model AIC:", AIC(logit_model_step), "\n")
870
871 # =====
872 # 13. COMPARING MODELS WITH ORIGINAL VS RECODED EDUCATION
873 # =====
874
875 cat("\n=== COMPARISON: Original Education vs Recoded Education ===\n")
876
877 # Fit model with original education for comparison
878 logit_model_original_edu <- glm(
879   happiness_binary ~ religion + tolerance + authority_respect +
880   economical_satisfaction + gender + age + region + education,
881   data = df_model,
882   family = binomial(link = "logit")
883 )
884
885 cat("\nModel with Original Education AIC:", AIC(logit_model_original_edu), "\n")
886 cat("Model with Recoded Education AIC:", AIC(logit_model_full), "\n")
887
888 # Full model predictions (you already have these)
889 df_model$pred_prob_full <- predict(logit_model_full, type = "response")
890
891 # Stepwise model predictions (NEW)
892 df_model$pred_prob_step <- predict(logit_model_step, type = "response")
893 df_model$pred_class_step <- ifelse(df_model$pred_prob_step > 0.5, 1, 0)
894 # Full model
895 conf_matrix_full <- table(Actual = df_model$happiness_binary,
896   Predicted = df_model$pred_class)
897
898 # Stepwise model
899 conf_matrix_step <- table(Actual = df_model$happiness_binary,
900   Predicted = df_model$pred_class_step)
901
902 print("Full Model:")
903 print(conf_matrix_full)
904
905 print("Stepwise Model:")
906 print(conf_matrix_step)
907
908 # Function to calculate metrics
909 calculate_metrics <- function(conf_matrix) {
910   accuracy <- sum(diag(conf_matrix)) / sum(conf_matrix)
911   sensitivity <- conf_matrix[2,2] / sum(conf_matrix[2,])
912   specificity <- conf_matrix[1,1] / sum(conf_matrix[1,])
913
914   return(list(accuracy = accuracy,
915     sensitivity = sensitivity,
916     specificity = specificity))
917 }
918
919 metrics_full <- calculate_metrics(conf_matrix_full)
920 metrics_step <- calculate_metrics(conf_matrix_step)
921
922 cat("\n=== FULL MODEL ===\n")

```

```

922 cat("\n=== FULL MODEL ===\n")
923 print(metrics_full)
924
925 cat("\n=== STEPWISE MODEL ===\n")
926 print(metrics_step)
927
928
929
930 # ROC for full model
931 roc_full <- roc(df_model$happiness_binary, df_model$pred_prob_full)
932 auc_full <- auc(roc_full)
933
934 # ROC for stepwise model
935 roc_step <- roc(df_model$happiness_binary, df_model$pred_prob_step)
936 auc_step <- auc(roc_step)
937
938 cat("\nFull Model AUC:", round(auc_full, 3), "\n")
939 cat("Stepwise Model AUC:", round(auc_step, 3), "\n")
940
941 # Plot both ROC curves together
942 library(ggplot2)
943 roc_comparison <- data.frame(
944   Model = rep(c("Full Model", "Stepwise Model"),
945     c(length(roc_full$specificities), length(roc_step$specificities))),
946   Specificity = c(roc_full$specificities, roc_step$specificities),
947   Sensitivity = c(roc_full$sensitivities, roc_step$sensitivities)
948 )
949
950 ggplot(roc_comparison, aes(x = 1 - Specificity, y = Sensitivity, color = Model)) +
951   geom_line(size = 1.2) +
952   geom_abline(intercept = 0, slope = 1, linetype = "dashed", color = "gray") +
953   labs(title = "ROC Curve Comparison: Full vs Stepwise Model",
954     x = "False Positive Rate (1 - Specificity)",
955     y = "True Positive Rate (Sensitivity)") +
956   theme_minimal() +
957   scale_color_manual(values = c("Full Model" = "darkblue",
958     "Stepwise Model" = "darkred"))
959
960 # 12b. ROC CURVE FOR STEPWISE MODEL
961
962 # Create ROC object for stepwise model
963 roc_obj_step <- roc(df_model$happiness_binary, df_model$pred_prob_step)
964
965 # Calculate AUC for stepwise model
966 auc_value_step <- round(as.numeric(auc(roc_obj_step)), 3)
967 cat("\n=== Stepwise Model AUC ===\n")
968 cat("AUC:", auc_value_step, "\n")
969
970 # Plot ROC curve for stepwise model
971 ggroc(roc_obj_step, legacy.axes = TRUE, size = 1.4, colour = "darkred") +
972   geom_abline(intercept = 0, slope = 1,
973     linetype = "dashed", colour = "gray50", size = 1) +
974   labs(
975     title = paste("ROC Curve - Stepwise Model - AUC =", auc_value_step),
976     subtitle = ifelse(auc_value_step >= 0.8, "Excellent discrimination",
977       ifelse(auc_value_step >= 0.7, "Acceptable discrimination",
978         "Poor discrimination")),
979     x = "False Positive Rate (1 - Specificity)",
980     y = "True Positive Rate (Sensitivity)"
981   ) +
982   theme_minimal(base_size = 14) +
983   theme(plot.title = element_text(face = "bold"))
984
985 # 12c. SENSITIVITY AND SPECIFICITY BY THRESHOLD - STEPWISE MODEL
986
987 #
988
989 cat("\n=== Sensitivity and Specificity by Threshold - Stepwise Model ===\n")
990
991 # Extract sensitivity and specificity at different thresholds
992 roc_data_step <- coords(roc_obj_step, "all",
993   ret = c("threshold", "sensitivity", "specificity"),
994   transpose = FALSE) %>%
995   as.data.frame() %>%
996   pivot_longer(cols = c(sensitivity, specificity),
997     names_to = "metric",
998     values_to = "value")
999
1000 # Plot sensitivity and specificity by threshold for stepwise model
1001 ggplot(roc_data_step, aes(x = threshold, y = value, color = metric)) +
1002   geom_line(size = 1.2) +
1003   scale_color_manual(
1004     values = c("sensitivity" = "darkred", "specificity" = "darkblue"),
1005     labels = c("Sensitivity", "Specificity")
1006   ) +
1007   labs(
1008     title = "Stepwise Model: Sensitivity and Specificity by Threshold",
1009     x = "Threshold (Predicted Probability)",
1010     y = "Rate",
1011     color = ""
1012   ) +
1013   theme_minimal(base_size = 14) +
1014   theme(legend.position = "bottom")
1015
1016 #
1017 # 12d. OPTIMAL THRESHOLD FOR STEPWISE MODEL
1018
1019 #

```

```

1019 # =====
1020
1021 # Find optimal threshold using Youden's index
1022 optimal_coords_step <- coords(roc_obj_step, "best",
1023                               ret = c("threshold", "sensitivity", "specificity"),
1024                               best.method = "youden")
1025 cat("\n=== Optimal Threshold - Stepwise Model (Youden's Index) ===\n")
1026 print(optimal_coords_step)
1027
1028 # Apply optimal threshold for stepwise model
1029 optimal_threshold_step <- optimal_coords_step$threshold
1030 df_model$pred_class_step_optimal <- ifelse(df_model$pred_prob_step > optimal_threshold_step, 1, 0)
1031
1032 # Confusion matrix with optimal threshold
1033 cat("\n=== Confusion Matrix - Stepwise Model (Optimal Threshold) ===\n")
1034 conf_matrix_step_opt <- table(Actual = df_model$happiness_binary,
1035                               Predicted = df_model$pred_class_step_optimal)
1036 print(conf_matrix_step_opt)
1037
1038 # Calculate metrics at optimal threshold
1039 accuracy_step_opt <- sum(diag(conf_matrix_step_opt)) / sum(conf_matrix_step_opt)
1040 sensitivity_step_opt <- conf_matrix_step_opt[2,2] / sum(conf_matrix_step_opt[2,])
1041 specificity_step_opt <- conf_matrix_step_opt[1,1] / sum(conf_matrix_step_opt[,1])
1042
1043 cat("\nStepwise Model - Optimal Threshold (", round(optimal_threshold_step, 3), "):\n")
1044 cat("Accuracy:", round(accuracy_step_opt, 3), "\n")
1045 cat("Sensitivity:", round(sensitivity_step_opt, 3), "\n")
1046 cat("Specificity:", round(specificity_step_opt, 3), "\n")
1047
1048 # =====
1049 # 12e. COMPARE FULL MODEL VS STEPWISE MODEL
1050 # =====
1051
1052 # Plot ROC curves together for comparison
1053 roc_comparison_data <- rbind(
1054   data.frame(
1055     Model = "Full Model",
1056     FPR = 1 - roc_obj$specificities,
1057     TPR = roc_obj$sensitivities
1058   ),
1059   data.frame(
1060     Model = "Stepwise Model",
1061     FPR = 1 - roc_obj_step$specificities,
1062     TPR = roc_obj_step$sensitivities
1063   )
1064 )
1065
1066 ggplot(roc_comparison_data, aes(x = FPR, y = TPR, color = Model)) +
1067   geom_line(size = 1.2) +
1068   geom_abline(intercept = 0, slope = 1, linetype = "dashed", color = "gray50") +
1069   scale_color_manual(values = c("Full Model" = "darkblue", "Stepwise Model" = "darkred")) +
1070   labs(
1071     title = "ROC Curve Comparison: Full Model vs Stepwise Model",
1072     subtitle = paste0("Full AUC = ", auc_value, " | Stepwise AUC = ", auc_value_step),
1073     x = "False Positive Rate (1 - Specificity)",
1074     y = "True Positive Rate (Sensitivity)"
1075   ) +
1076   theme_minimal(base_size = 14) +
1077   theme(plot.title = element_text(face = "bold"),
1078         legend.position = "bottom")
1079
1080 cat("\n=== RECOMMENDATION ===\n")
1081 if (abs(auc_value - auc_value_step) < 0.02 && AIC(logit_model_step) < AIC(logit_model_full)) {
1082   cat("CHOOSE STEPWISE MODEL:\n")
1083   cat("  - Similar performance (AUC difference < 0.02)\n")
1084   cat("  - Lower AIC (better parsimony)\n")
1085   cat("  - Fewer predictors (simpler model)\n")
1086 } else if (auc_value - auc_value_step > 0.02) {
1087   cat("CONSIDER KEEPING FULL MODEL:\n")
1088   cat("  - Full model has notably better discrimination\n")
1089   cat("  - AUC difference: ", round(auc_value - auc_value_step, 3), "\n")
1090 } else {
1091   cat("Both models perform similarly. Choice depends on your priorities.\n")
1092 }
1093 #####
1094 #                               END OF THE CODE                               #
1095 #####
1096

```